

Strategy Evidence Data Architecture (2)

CONFIDENTIAL

Strategy, Evidence & Data Architecture

Internal Master Planning Document

Document 3

Purpose: Define the strategic opportunities available within the business, the evidence required to evaluate them, and the datasets that must exist within the RAG to support meaningful analysis.

Parent Documents:

Document 1 — Company Blueprint

Document 1.5 — Business Dependency Map

Document 2 — Consulting Engagement Letter

1. How This Document Should Be Used

This document defines the intended solution space, not the expected answer.

A participant should be able to:

Discover a business problem.

Form a hypothesis.

Find supporting evidence.

Combine multiple datasets.

Develop a strategy.

Estimate its impact.

Identify trade-offs.

Defend the recommendation.

No strategy should be directly revealed by a single dataset.

2. Strategic Solution Universe

We will build the case around 10 primary strategic pathways.

These are broad enough to permit unconventional solutions while remaining understandable to engineering students.

S-01 — Marketing Efficiency

Core Question: Is the company spending its marketing budget in the right places?

Dependencies: D-001 → D-006, D-035 → D-036

S-02 — Customer Retention

Core Question: Are we acquiring customers successfully but failing to capture their second purchase?

Dependencies: D-011 → D-015, D-024 → D-026

S-03 — B2B Expansion

Core Question: Is the company underutilising its corporate sales opportunity?

Dependencies: D-031 → D-034

S-04 — Product Mix Optimisation

Core Question: Are we selling the products that create the best combination of revenue and profit?

Dependencies: D-007 → D-010, D-017 → D-023

S-05 — Pricing Strategy

Core Question: Are our current prices leaving revenue or margin on the table?

Dependencies: D-004 → D-010

S-06 — Average Order Value

Core Question: Can we generate more revenue from each existing transaction?

Dependencies: D-007 → D-010

S-07 — Inventory Optimisation

Core Question: Are we losing sales because the right products aren't available at the right time?

Dependencies: D-016 → D-023

S-08 — Channel Optimisation

Core Question: Are our different sales channels being managed effectively?

Dependencies: D-001 → D-010, D-035 → D-037

S-09 — Customer Experience

Core Question: Could improving the post-purchase experience generate additional growth?

Dependencies: D-011 → D-015, D-019 → D-026

S-10 — Regional Growth

Core Question: Are some regions or cities significantly outperforming others?

Dependencies: D-002 → D-006, D-016 → D-020

3. Secondary / Hybrid Strategies

This is extremely important.

We should not force participants to choose one of the ten strategies.

The strongest teams should be able to combine them.

For example:

Retention + Product Mix

Identify customers who bought earbuds but never purchased chargers or power banks.

→ targeted cross-selling

→ increased repeat purchases

→ higher AOV

Marketing + Inventory

Identify campaigns generating demand for products frequently experiencing stock-outs.

→ redirect marketing

→ reduce wasted acquisition spend

→ improve conversion

B2B + Product Mix

Identify products with high consumer margins that can also be sold in corporate bundles.

→ B2B expansion

→ higher-margin revenue

Pricing + Inventory

Use selective discounts to clear slow-moving inventory while protecting margins on high-demand products.

→ working capital release

→ inventory capacity

→ additional sales

Customer Experience + Retention

Identify products generating disproportionate complaints and returns.

→ resolve product/service issues

→ improve ratings

→ improve repeat purchase

4. Evidence Architecture

Every major strategy should require multiple evidence sources.

This directly enforces Cardinal Rule #3:

No single dataset should reveal the best solution on its own.

5. Core Dataset Architecture

Now we get to the part that matters most.

We should build approximately 20–25 core datasets, rather than dozens of unnecessary files.

A. SALES DATA

DS-01 — Transaction-Level Sales

Grain: One row = one order line

Fields: Order ID, Order Date, Customer ID, Product ID, Quantity, Selling Price, Discount, Revenue, Channel, Region, Warehouse, Return Flag.

Supports: S-01, S-04, S-05, S-06, S-08, S-10

DS-02 — Channel Performance

Grain: Monthly channel-level

Fields: Revenue, Orders, Units, AOV, Conversion, Returns, Margin, Acquisition Cost.

Supports: S-01, S-08

DS-03 — Regional Sales

Grain: Monthly / region

Fields: Revenue, Orders, Customers, AOV, Growth, Product mix.

Supports: S-08, S-10

B. MARKETING DATA

DS-04 — Campaign Performance

Fields: Campaign ID, Channel, Audience, Spend, Impressions, Clicks, CTR, Conversions, Revenue, CAC, ROAS.

Supports: S-01, S-08, S-10

DS-05 — Website Analytics

Fields: Sessions, Unique Visitors, Product Views, Add-to-Cart, Checkout, Purchases, Conversion Rate, Traffic Source.

Supports: S-01, S-05, S-06

DS-06 — Marketing Budget

Grain: Monthly

Fields: Allocated Budget, Actual Spend, Remaining Budget, Department, Campaign, Channel.

Supports: S-01 and financial feasibility

C. CUSTOMER DATA

DS-07 — Customer Master / CRM

Fields: Customer ID, Segment, Region, Acquisition Channel, First Purchase, Last Purchase, Total Orders, Total Revenue, Customer Value, Customer Status.

Supports: S-02, S-06, S-10

DS-08 — Customer Purchase History

This is separate from the transaction table for analytical convenience.

Fields: Customer, Purchase Number, Product, Purchase Date, Order Value, Days Since Previous Purchase.

Supports: S-02, S-06

DS-09 — Customer Feedback

Fields: Customer ID, Product, Rating, NPS, Feedback Category, Sentiment, Date.

Supports: S-02, S-09

DS-10 — Customer Support & Warranty

Fields: Ticket ID, Customer, Product, Issue Type, Response Time, Resolution Time, Resolution Status, Warranty Claim, Satisfaction.

Supports: S-02, S-09

D. PRODUCT DATA

DS-11 — Product Master

Fields: Product ID, Product, Category, Launch Date, List Price, Cost, Margin, Target Segment, Status.

Supports: S-04, S-05, S-06

DS-12 — Product Performance

Grain: Monthly SKU-level

Fields: Units Sold, Revenue, Margin, Returns, Rating, Stock-outs, Repeat Purchase Association.

Supports: S-02, S-04, S-05

DS-13 — Product Bundle / Basket Analysis

This dataset is particularly important for unconventional strategies.

Fields: Product A, Product B, Co-purchase Frequency, Customer Segment, Revenue, Margin.

Supports: S-02, S-04, S-06

E. INVENTORY & OPERATIONS

DS-14 — Inventory Snapshot

Fields: SKU, Warehouse, Opening Stock, Receipts, Units Sold, Closing Stock, Days of Inventory, Stock-out Days.

Supports: S-04, S-07, S-10

DS-15 — Warehouse Performance

Fields: Warehouse, Region Served, Orders, Fulfilment Rate, Processing Time, Stock-outs, Capacity Utilisation.

Supports: S-07, S-10

DS-16 — Supplier Performance

Fields: Supplier, Product, Lead Time, Promised Delivery, Actual Delivery, Defect Rate, Cost, Reliability.

Supports: S-07

DS-17 — Delivery & Fulfilment

Fields: Order, Warehouse, Dispatch Time, Delivery Time, Delivery Status, Delay, Region, Customer Rating.

Supports: S-07, S-09, S-10

F. FINANCE

DS-18 — Monthly P&L

Fields: Revenue, COGS, Gross Profit, Marketing Expense, Operations Expense, Payroll, Other Expenses, Operating Profit.

Supports: ALL STRATEGIES

DS-19 — Cost Structure

Fields: Cost Category, Fixed/Variable, Monthly Cost, Department, Cost per Order, Cost per Unit.

Supports: Financial feasibility of ALL strategies

DS-20 — Budget Allocation

Fields: Department, Approved Budget, Actual Spend, Remaining, Proposed Allocation.

Supports: All strategies

G. B2B

DS-21 — B2B Sales Pipeline

Fields: Lead, Company, Industry, Deal Size, Product, Stage, Probability, Expected Revenue, Sales Rep, Expected Close Date.

Supports: S-03

DS-22 — B2B Historical Performance

Fields: Account, Order, Deal Size, Margin, Sales Cycle, Repeat Order, Product Mix.

Supports: S-03, S-04

H. MARKET / COMPETITIVE DATA

DS-23 — Competitor & Market Pricing

Fields: Competitor, Product Category, Comparable Product, Price, Promotion, Rating, Feature Positioning, Market Position.

Supports: S-05, S-08, S-10

I. FORECASTING

DS-24 — Demand Forecast

Fields: Product, Region, Month, Forecast Demand, Actual Demand, Forecast Error, Confidence.

Supports: S-07, S-10

J. CUSTOMER COHORTS

DS-25 — Cohort Performance

Fields: Acquisition Month, Acquisition Channel, Customer Segment, First Purchase, Repeat Purchase, 30/60/90-Day Retention, Revenue, CLV.

Supports: S-01, S-02, S-06

6. Dataset Relationship Architecture

These datasets must not be generated independently.

The primary relationship structure should be:

This means DS-01 cannot simply be invented after DS-04.

The datasets need a common underlying business model.

7. The Data Generation Rule

This is going to be one of our most important internal rules.

We should not generate:

We should generate:

For example:

If marketing spend increases:

→ Traffic should respond.

If traffic increases:

→ Orders should respond according to conversion.

If orders increase:

→ Inventory should decrease.

If inventory gets too low:

→ Stock-outs should occur.

If stock-outs occur:

→ Lost sales and customer dissatisfaction should increase.

That is the level of interconnectedness we need.

8. Data Difficulty Design

We specifically want to reward both conventional and unconventional strategies.

Therefore:

Conventional Strategy

Should be discoverable through relatively obvious relationships.

Example:

Unconventional Strategy

Should require combining seemingly unrelated information.

Example:

That could lead to:

Targeted post-purchase cross-selling

rather than simply "increase advertising."

9. What We Should NOT Do

We should avoid:

✗ Artificially hiding information

We shouldn't make the answer difficult by withholding obvious information.

✗ Making datasets unnecessarily huge

Large datasets aren't automatically good datasets.

✗ Creating random noise

Noise should exist only where it reflects realistic business uncertainty.

✗ Creating contradictory numbers

Ambiguity is good. Contradiction is not.

✗ Making one strategy obviously dominant

Every strong strategy should have a trade-off.

10. Dataset Priority

Because we're working under a four-day deadline, not all 25 datasets have equal priority.

Tier 1 — Absolutely Essential

DS-01 Transaction Sales

DS-04 Campaign Performance

DS-05 Website Analytics

DS-07 Customer CRM

DS-08 Purchase History

DS-09 Customer Feedback

DS-11 Product Master

DS-12 Product Performance

DS-14 Inventory

DS-18 P&L

DS-20 Budget

DS-21 B2B Pipeline
DS-23 Competitor Pricing
DS-24 Demand Forecast
DS-25 Cohorts

Tier 2 — Important

DS-02, DS-03, DS-06, DS-10, DS-13, DS-15, DS-16, DS-17, DS-19, DS-22

Tier 3 — Optional

Any additional datasets that emerge during QA.

We do not generate Tier 3 until Tier 1 is rock solid.

11. Cardinal Rule Check

This architecture satisfies our Cardinal Rules because:

Rule 1: Every major dataset exists to support a business decision.

Rule 2: Every major strategy has evidence pathways.

Rule 3: No single dataset reveals the solution.

Rule 4: Multiple interacting inefficiencies are built into the architecture.

Rule 5: Reasoning is required to connect evidence.

Rule 6: Disruptions can enter multiple systems without destroying the case.

Rule 7: Each dataset has a defined business purpose.

Rule 8: Numbers will be generated from interconnected relationships.

Rule 9: KPIs have upstream and downstream dependencies.

12. The Actual Build Order

This is the part I want the tech team to see.

We should not generate the datasets in numerical order.

We build the underlying entities first:

Product Master

Customer Master

Supplier Master

Warehouse Master

Channel Master

Region Master

↓

Transaction Engine

↓

Marketing

Inventory

Operations

Customer Behaviour

B2B

Finance

Market Data



Derived KPIs



RAG Documents

This is much safer.

Because if we generate the P&L first and then invent sales data that "looks right," we're going to spend the next three days fixing contradictions.

13. Final Architecture

The whole competition now looks like:



STRATEGIC OBJECTIVE
Identify the most impactful growth opportunity within the business and recommend a practical, evidence-backed strategy c

Potential Approaches
Reallocate campaign budgets.
Reduce low-performing campaigns.
Increase high-performing channels.
Improve

Potential Approaches
Retention campaigns
Cross-selling
Product recommendations
Warranty engagement
Loyalty incentives
A

Potential Approaches
Dedicated B2B acquisition
Corporate bundles
Institutional partnerships
Bulk pricing
Corporate gifting
A

Potential Approaches
Promote high-margin products.
Reduce emphasis on weak products.
Bundle complementary products.
Cross-se

Potential Approaches
Selective price increases
Targeted discounts
Product-specific pricing
Bundled pricing
Channel-specific pric

Potential Approaches Product bundles Cross-selling Quantity incentives Accessories Premium upgrades	Evidence Required
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Potential Approaches
Reallocate inventory between warehouses.
Prioritise high-demand products.
Reduce slow-moving stock.
Improv

Potential Approaches
Shift channel investment.
Improve website conversion.
Reduce marketplace dependence.
Expand offline retail se

Potential Approaches
Reduce resolution time.
Improve warranty handling.
Address high-return products.
Improve delivery experience.

Potential Approaches
Regional marketing allocation
Localised promotions
Retail expansion
Regional inventory allocation
Regiona

